

A group of people are gathered around a table, looking at a laptop screen. A woman on the left is looking intently at the screen. A woman on the right is smiling and looking towards the camera. A third person is partially visible on the right, holding a pen. The laptop screen displays a web application with various charts and data. There are also some papers and a cup on the table.

Java fra scratch

Program

- 08:30 Lidt om læring, AI og fremtiden som programmør
- 09:00 Hvad er Java
- 09:30 Installering af Java og IntelliJ
- 10:00 Pause
- 10:15 Første program
- 11:15 Opsamling

A group of people are gathered around a table in a meeting. A woman on the left is looking at a laptop screen. A woman on the right is smiling and looking towards the camera. A third person is partially visible on the right, holding a pen. The laptop screen displays a software interface with various charts and data. A white text box is overlaid in the center of the image.

Hardware og software

Computeren er en **standardmaskine**, der kan udføre mange **forskellige opgaver**

Hardware

- CPUen beregner og udfører instrukser
- RAM er midlertidig arbejdshukommelse
- Harddisk er permanent hukommelse - filer

Software

- **Operativsystem** styrer computeren
 - fx Windows, MacOS, Android, iOS
- **Drivere** får computeren til at kommunikere med hardware
 - fx keyboard, printer, skærm
- **Applikationer** er det, vi bruger computeren til
 - fx Word, Photoshop, browsere, spil

A top-down view of a person sitting on a yellow armchair, typing on a silver laptop. The person is wearing a red and black jacket with camouflage patterns on the sleeves and a silver bracelet on their left wrist. In the foreground, a tablet displays a code editor with syntax-highlighted code. To the left of the laptop, an O'Reilly book titled 'Web Craft' by Jeremy M. Jones is visible, featuring a detailed illustration of a dragon on its cover.

Programmieringssprog

مرحبا
(marHaba)



ARABIC

Здравствуйте
(zdrastvooite)



RUSSIAN

Hola
(ola)



SPANISH

Hallo
(halo)



GERMAN

Bonjour
(boñ-zhoor)



FRENCH

Ciao
(chow)



ITALIAN

Olá
(o-laa)



PORTUGUESE

नमस्ते
(namastey)



HINDI

Hello World in 30 different languages

C

```
#include

int main(void)
{
    puts("Hello, world!");
}
```

Matlab

```
disp('Hello, world!')
```

Pascal

```
WriteLn('Hello, world!');
```

Go

```
println('Hello, world!');
```

F#

```
printfn "Hello World"
```

Lisp

```
(print "Hello world")
```

C#

```
Console.WriteLine("Hello, world!");
```

Ruby

```
puts "Hello World!"
```

Java

```
System.out.println("Hello World!");
```

JavaScript

```
console.log 'Hello, world!'
```

C++

```
#include

int main()
{
    std::cout << "Hello, world!";
    return 0;
}
```

CoffeeScript

```
console.log 'Hello, world!'
```

Python

```
print('Hello, world!')
```

PHP

```
echo "Hello World!";
```

Algol

```
BEGIN DISPLAY("HELLO WORLD!") END.
```

Delphi

```
program HelloWorld;
begin
    WriteLn('Hello, world!');
end.
```

Assembly

```
global _main
extern _printf

section .text
_main:
    push    message
    call    _printf
    add     esp, 4
    ret
message:
    db 'Hello, World', 10, 0
```

Pascal

```
program HelloWorld(output);
begin
    Write('Hello, world!')
end.
```

Perl

```
print "Hello, World!\n";
```

Tcl

```
puts "Hello World!"
```

Cobol

```
IDENTIFICATION DIVISION.
PROGRAM-ID. hello-world.
PROCEDURE DIVISION.
    DISPLAY "Hello, world!"
```

Dart

```
main() {
    print('Hello World!');
}
```

Kotlin

```
fun main(args: Array<String>) {
    println("Hello World!")
}
```

TypeScript

```
console.log 'Hello, world!'
```

Scala

```
object HelloWorld extends App {
    println("Hello, world!")
}
```

Haskell

```
module Main where

main :: IO ()
main = putStrLn "Hello, World!"
```

R

```
cat("Hello world\n")
```

Swift

```
println('Hello, world!');
```

HTML

```
Hello world
```

Fortran

```
program helloworld
    print *, "Hello world!"
end program helloworld
```

May 2025	May 2024	Change	Programming Language		Ratings
1	1			Python	25.35%
2	3	▲		C++	9.94%
3	2	▼		C	9.71%
4	4			Java	9.31%
5	5			C#	4.22%
6	6			JavaScript	3.68%
7	8	▲		Go	2.70%
8	7	▼		Visual Basic	2.62%
9	11	▲		Delphi/Object Pascal	2.29%

Java er et **general purpose programmeringssprog**



A photograph of two men in a modern office setting. The man on the left has a full beard and is wearing a dark jacket over a plaid shirt. The man on the right is wearing a plaid shirt and is looking at a laptop. They are sitting at a desk with multiple computer monitors displaying code. A large window in the background shows a cityscape. The word "Syntaks" is overlaid in a white serif font on a black rectangular background.

Syntaks

Syntax - regler for rækkefølge af ord

"spiser en sandwich jeg"

Udsagnsled + **grundled** + **genstandsled**

eller

"jeg spiser en sandwich"

grundled + **udsagnsled** + **genstandsled**

Korrekt

```
int age = 18;
```

Forkert

```
age = 18 int;
```

```
int = 18 age;
```

```
INT age = 18;
```


Korrekt

```
System.out.println("Hello, World!");
```

Forkert

```
System.out.println Hello, World!;
```

```
System.out.println(Hello, World!");
```

```
System.out println("Hello, World!");
```

Korrekt

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Forkert

```
public class HelloWorld {}  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }
```

```
public class HelloWorld {  
    public static void (String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

```
public HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Første program



1. Lav en fil, der hedder HelloWorld.java
2. Skriv følgende kode i filen og gem den:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```


3. Compile med

```
javac HelloWorld.java
```

4. Kør programmet med

```
java HelloWorld
```

5. Hvad sker der?

Kompilering

Compileren **javac** oversætter kildekode **HelloWorld.java** til bytecode **HelloWorld.class**

```
javac HelloWorld.java
```

Java bytecode

er en binær fil, der ender på `.class`

Whitespace

- mellemrum, tabulatorer og linjeskift
- **ikke** vigtigt for computeren

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

er **korrekt** syntaks

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

er **også korrekt** syntaks

```
public class HelloWorld { public static void main(String[] args) { System.out.println("Hello, World!");
```

er **også korrekt**

Hvorfor synes I vi skal bruge **indrykning**?



Tre ting du tager med fra fra i dag?